Jialin Huang

Software Engineer

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Open to relocation anywhere · Mandarin (native), English (professional)

Software engineer, eight years. Most recently on a multi-tenant security SaaS, writing Go and Node services for identity and access: claims-based RBAC on Cognito and GraphQL, SAML and social sign-in, and a Go audit pipeline for auth events. On the build side I moved the monorepo to pnpm (installs 3-10x faster) and cut two Docker images by about 80%. Frontend background; React and TypeScript daily.

SKILLS

Languages: TypeScript, Go

Backend: NestJS, Apollo GraphQL, Gin, Elasticsearch, Node.js, GraphQL, SignalR, Firebase Auth, C#/.NET, Python/Flask

Security & Domain: RBAC, SAML, OAuth2/OIDC, token claims, audit logging, Zero Trust access, multi-tenant SaaS

Infrastructure: PostgreSQL, Redis, Docker, Kubernetes, Jenkins, AWS Lambda, AWS Cognito, Playwright, Firebase, GCP

Frontend: React, Next.js, Remix, Vue, Canvas, WebSocket, Google Analytics, Tailwind CSS, Emotion, Redux

EXPERIENCE

MammothCyber

Senior Software Engineer · Cybersecurity · Taipei (Hybrid) · Apr 2025 - Jul 2026

Full-stack across the web portal of a multi-tenant enterprise security platform, a Chromium-based secure browser, and Go services; identity & access domain (SAML, OIDC/JWT, RBAC, audit, policy config)

Identity, Access & Audit

Worked across the auth stack (Go + Node): added tenant-level forced re-authentication (SAML), and threaded an OS / device-posture field across the login-policy service, the Cognito audit pipeline, and the browser and extension clients

Claims-based RBAC end to end: permissions computed in the Cognito PreTokenGeneration Lambda and compressed into the id_token, enforced by a Casbin guard behind a GraphQL @rbac directive

Auth-event audit pipeline in Go: Cognito CloudTrail events polled from SQS, classified (password / MFA / device failures), enriched from PostgreSQL with an LRU cache, shipped to Logstash

Added Google / Apple social login (OAuth) to Cognito alongside the existing SAML, reworking the Lambda triggers that map each social user to its tenant

Platform & Build Infrastructure

Drove a monorepo dependency cleanup on npm: root overrides and targeted upgrades (Express v5, Apollo Server v5, ESLint v9) cut vulnerabilities 74 → 13 (-82%), clearing every high & critical, and stray local packages 22 → 0

npm to pnpm: Migrated the monorepo to pnpm workspaces with a staged rollout (shamefully-hoist bridge, declare every phantom dep, flip to strict isolation), cutting installs 3-10x and making per-worktree installs cheap so parallel AI agents can each build in their own git worktree

Cut two Docker images ~80% in three steps: per-workspace pnpm deploy --prod instead of copying the root node_modules, misdeclared prod deps fixed, then both servers bundled (Remix ssr.noExternal + esbuild, NestJS nest build --webpack) so the runtime image ships no node_modules. Remix web app 1.88 GB → 264 MB, NestJS API 1.95 GB → 490 MB

Wrote the integration tests in Playwright, running pages in parallel

Policy Configuration Console

Built the policy-configuration backend in NestJS: rule and content-profile APIs, RBAC-gated, with audit logging and an N+1 fix that collapsed a multi-call save into a single request

Built the rules and content-profile editor in Remix, type-safe end to end with reactive validation

Secure Browser & Extension (Chromium)

Shipped a login / session-visibility feature across every client of a Chromium-based enterprise browser, extending each platform's native bridge in C++ (Android / Desktop) and Objective-C++ (iOS), plus a Go fetch backend and a React UI

Console Migration to Remix

Migrated console pages onto Remix (React 18, Mantine, TanStack Query/Table, Vite) off an earlier Vue / Next.js stack, page by page with new and old running side by side

Bunastar

Frontend Engineer · Gaming · Taipei (Hybrid) · Nov 2024 - Apr 2025

Real-time multiplayer card game for the Southeast Asian market; custom Canvas engine and WebSocket sync

Southeast Asian Gaming Platform - Pok Deng

Card game logic: dealing, turn order, scoring, round settlement

Real-time card sync over WebSocket (client side)

Canvas rendering engine, optimized with sprite sheets

Cross-domain communication via postMessage API

Confetti

Full Stack Engineer · Gaming · Taipei (Hybrid) · Dec 2022 - Oct 2024

confetti.studio

Y2K landing page

Gaming Website Rewrite

Rewrote the platform API in Node.js

Rewrote the website in Next.js, React and Tailwind CSS

SCMP

Frontend Engineer · Media · Hong Kong (Remote) · Oct 2021 - Nov 2022

South China Morning Post: high-traffic international news platform, delivered fully remote

scmp.com

Improved Core Web Vitals (LCP, CLS), Google's SEO ranking signals, on scmp.com, a high-traffic international news site: fixed the homepage hero's layout shift, lazy-loaded below-fold images, and tuned SSR data prefetch

Applied GDPR right-to-erasure in the comment system: surfaced a gdpr_erased flag through the GraphQL layer so erased users' names render as Anonymous

Maintained the CMS-driven content rendering pipeline (Node.js, GraphQL)

Built the site-wide tracking plan across sections: GA events through the GTM dataLayer for traffic, and Mixpanel via SDK keyed on the reader's uuid for per-user product behavior

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Futuresai

Frontend Engineer · Finance · Taipei (Hybrid) · Nov 2020 - Oct 2021

Trading Dashboard + Admin + Auth

Monorepo rewrite with SignalR for real-time trading-data streaming

Firebase auth middleware with custom membership logic, plus SMS and LINE push notifications

futures-ai.com

Built a subscription-billing system with multiple tiers

Live trading strategy articles with graphical analysis

Dentalltw

Frontend Engineer · Healthcare · Taipei (Hybrid) · Jul 2018 - Nov 2020

Built a 0-to-1 clinic product: a desktop app for Taiwan and a React web app for the Vietnam market

Dentall Clinic Management

Built the desktop clinic app (UWP, C#/.NET) for registration, appointments, billing, and prescriptions, integrated with Taiwan's NHI (National Health Insurance) system

Built an SVG-based interactive dental chart for procedure documentation

Vietnam Market Expansion

Built the Vietnam web app (React, Redux): the clinic software localized, with multi-language support and SMS

CI/CD Pipeline

Set up CI/CD to build and push Docker images to GCR, using a Go service and GitLab scripts

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PROJECTS

Palimpsest · Angular, TypeScript, Bash, Claude Code

A boilerplate for running an AI coding agent under enforced rules: an identity file models the user, and shell hooks refuse any write that breaks a rule, instead of asking the model to comply.

Three governance speeds in one charter: a bottomline no in-session pressure can move, principles that shift only on new evidence, and a free drift layer.

An Angular dashboard comparing 11 open-source "AI identity" projects, built by cloning each repo and reading its config: 5 of the 11 enforce their constraints, the rest only declare them.

URL Shortener · Go, Gin, PostgreSQL, Redis, Kafka, ClickHouse

One redirect touches five systems: PostgreSQL for link mappings, Redis for live counters, Kafka for the click pipeline, ClickHouse for analytics, and a Go/Gin service tying them together.

Clicks flow through Kafka in 500-event / 2s batches across 4 writer goroutines, so analytics writes never block a redirect

Write-behind counter: redirects INCR in Redis, a 5s worker drains to PostgreSQL; reads merge the durable total with the live delta, streamed to a dashboard over Server-Sent Events

221b · Python, ChromaDB, sentence-transformers, MCP

RAG over my own writing: blog posts, AI-agent sessions, and config files in one corpus, queried in Chinese, answered in my English voice.

Chinese question → multilingual-e5 retrieval over ~4,000 chunks → bge cross-encoder rerank → answer grounded in my English writing.

Eval harness reports Recall@5 and MRR against ground-truth articles, so a model or prompt swap shows up as a number.

Game of Life · WebGL, TypeScript

Conway's Game of Life running on the GPU.

Simulated 770K+ cells at 60fps with a WebGL fragment-shader pipeline

Kubelens · Angular, TypeScript

If you have kubectl, you have Kubelens. Nothing to install.

Interactive graph of how Deployments, Pods, Services, and Ingresses connect; drag any resource into its own panel and run rollouts there.

Snapshot the whole cluster to browse offline.

Writing

N+1 isn't a database problem. It's a shape.
HTTP 1.1, 2, and 3: what broke and what fixed it
Five ways to reach a service in Kubernetes
A packet's walk through a VPC, in AWS
Stopping XSS and CSRF: who's responsible for what
Inside React's FiberWorkLoop
CSS rendering: layers, threads, and why transform beats width
from setState to pixels
SPA is not CSR: the 2026 rendering strategy map
Hiroki's Ark

EDUCATION

National Chengchi University (NCCU) · B.A. in Economics